



T-201-GSKI Data Structures

In-class exercise implementing a *map* abstract data type

Objective

The objective of the exercise is to get hands-on experience in implementing a map, and understanding the importance of a good hash-function and a constant time load factor

Logistics

- You can work in a **group of 1 - 2** as it is helpful to discuss the topics with others.

Description

Data structures and abstract data types are essential in programming. One of the most important of those is a *map*, or as you know it in Python, a *dictionary*. In this exercise you get a hands-on experience implementing a custom map (with the same interface as a Python dictionary!). We provide skeleton code (see on *Canvas* module for class).

Perform the following tasks in this order:

- Implementing the following incomplete methods in the *MyMap* class (file *my_map.py*):
 - `__set_item__`
 - `__get_item__`
 - `__del_item__`
- Test your implementation, including by running the *test_basic* test (file *main.py*).
- Now, uncomment the “Test Python’s dict” and “Test My dict” *test_map* calls, and run the main program. It shows the average access time when looking for an item in the map (and the number of items in each bucket).
 - What happens to the average access time as the the number of items (*n*) in the map grow?
 - How does the initial number of buckets (*N*) affect this?
 - Do you see the same effect when testing Python’s dict? If not, why do you think this is the case?
 - In *my_dict.py*, in function *h1* replace the *polynomial_rolling_hash(s)* call with the (now commented out) *len(s)* call. How does this affect performance, and why? **Undo your changes to *h1*.**

Once done, call a teaching assistant over to discuss your result!
- Implement the last remaining incomplete function (it should double the size of the *self._array* list and rehash all existing elements into it).
 - `__resize__`
- Now, uncomment the “Test Python’s dict with resizing” *test_map* call, and run the main program:
 - How does the access time behave now as *n* and *N* grows?
 - **Once done, call a teaching assistant over and discuss your result!**

Good luck! 😊